

Dataflow Analysis Complexity and Correctness

The termination of the worklist algorithm for dataflow analysis relies on two conditions: the dataflow lattice having finite height and the flow functions being monotonic. A flow function f is monotonic iff $\sigma_1 \sqsubseteq \sigma_2$ implies $f(\sigma_1) \sqsubseteq f(\sigma_2)$ for all σ_1, σ_2 .

The correctness of a dataflow analysis depends on the local soundness of its flow functions. For every program configuration c_i in the trace of a program P , a flow function f is locally sound iff $(P \vdash c_i \leadsto c_{i+1})$ and $\alpha(c_i) \sqsubseteq \sigma_{n_i}$ and $f[P[n_i]](\sigma_{n_i}) = \sigma_{n_{i+1}} \Rightarrow \alpha(c_{i+1}) \sqsubseteq \sigma_{n_{i+1}}$.

Exercises

These exercises prove properties of parity analysis. Assume the following:

- A lattice $(L, \sqsubseteq)$ where $L = \{\top, O, V, \perp\}$ and $\perp \sqsubseteq \{O, V\} \sqsubseteq \top, O \sqcup V = \top$
- An abstraction function $\alpha : \mathbb{Z} \mapsto L$, defined as follows:

$$\alpha(n) = \begin{cases} V & \text{when } n \text{ is an even integer } (n \in \{2k : k \in \mathbb{Z}\}) \\ O & \text{when } n \text{ is an odd integer } (n \in \{2k + 1 : k \in \mathbb{Z}\}) \end{cases}$$

- a flow function f_P
- initial dataflow analysis assumptions σ_0 , in this case σ_0 maps all variables' initial states to $\top$.

1. Disprove the local soundness of the incorrect flow function $f_P[a := b](\sigma) = \sigma[a \mapsto O]$

For the following question, consider the following flow function for parity analysis:

$$f_P[a := b * c](\sigma) = \begin{cases} \sigma[a \mapsto V] & \text{if } \sigma(b) = V \vee \sigma(c) = V \\ \sigma[a \mapsto \perp] & \text{if } \neg(\sigma(b) = V \vee \sigma(c) = V) \wedge (\sigma(b) = \perp \vee \sigma(c) = \perp) . \\ \dots & \text{other cases not specified} \end{cases}$$

2. While the provided case is mathematically correct (an even number multiplied by anything is still even), it is not monotonic. Prove it.

For the next two questions, use the following (correct) flow function for parity analysis:

$$f_P[a := b * c](\sigma) = \begin{cases} \sigma[a \mapsto \perp] & \text{if } \sigma(b) = \perp \vee \sigma(c) = \perp \\ \sigma[a \mapsto O] & \text{if } \sigma(b) = O \wedge \sigma(c) = O \\ \sigma[a \mapsto V] & \text{if } (\sigma(b) = V \wedge \sigma(c) \neq \perp) \vee (\sigma(b) \neq \perp \wedge \sigma(c) = V) \\ \sigma[a \mapsto \top] & \text{if } (\sigma(b) = \top \wedge \sigma(c) \notin \{V, \perp\}) \vee (\sigma(b) \notin \{V, \perp\} \wedge \sigma(c) = \top) \end{cases}$$

3. Prove the monotonicity of $f_P[a := b * c](\sigma)$ for the case $(\sigma(b) = V \wedge \sigma(c) \neq \perp) \vee (\sigma(b) \neq \perp \wedge \sigma(c) = V)$

4. Prove the local soundness of $f_P[a := b * c](\sigma)$